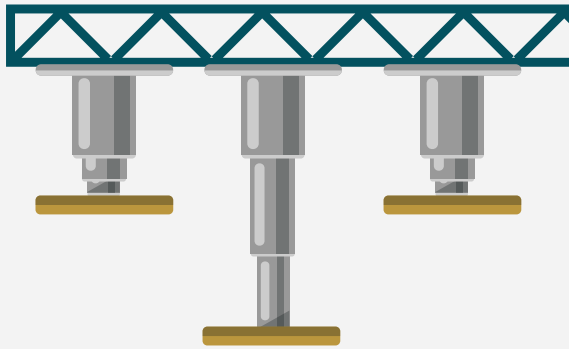


Monte Carlo
Analysis of a Multi-Machine Line

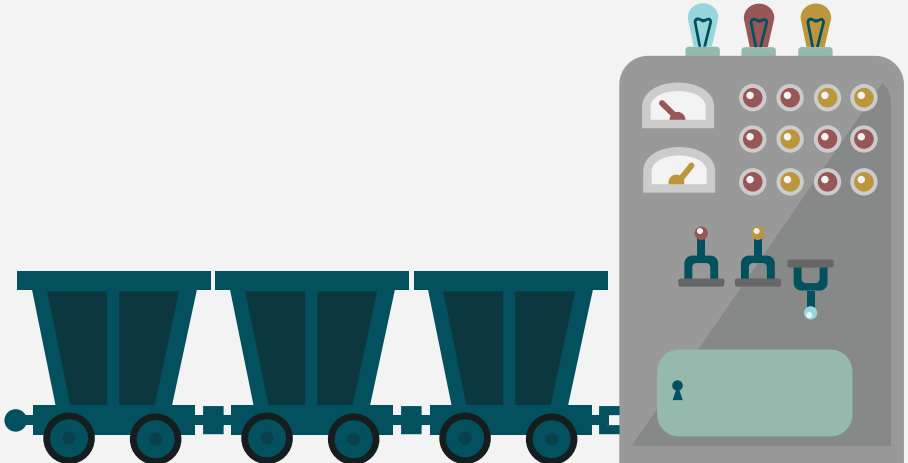


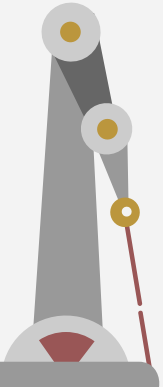
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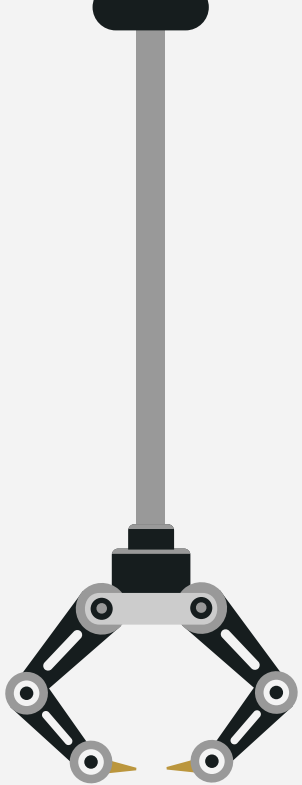
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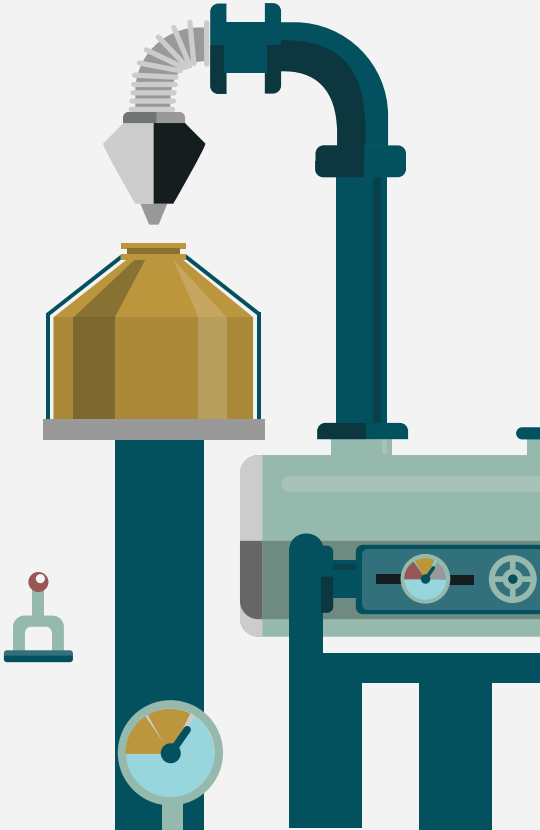






01

Recap of the Idea





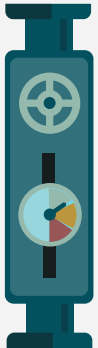
Keeping a 3-machine line on target

- A production line of three machines in series: M1 → M2 → M3.
- Machines fail at random and take a random time to repair; some units are scrapped.
- Buffers between machines absorb stoppages, but take space and hold inventory.
- Maintenance is either reactive (fix after a failure) or preventive (stop a machine every X hours).

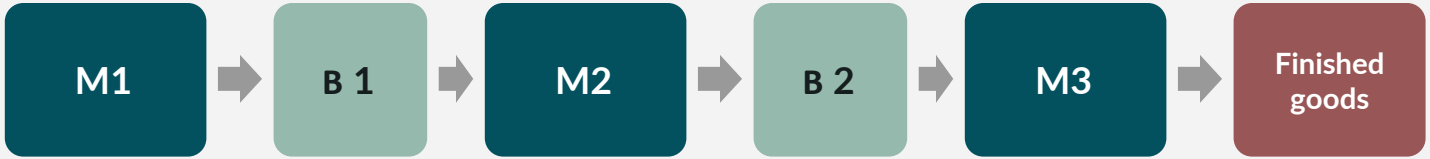
Lever 1: buffer size
B1 and B2 each set to 10, 20, 30, 40 or 50 units

Lever 2: maintenance policy
Reactive, or preventive every 100, 150, 200 or 250 h

Which combination meets demand most reliably?



Three machines, two buffers



Random failures and scrap

Lever 1 size 10 to 50

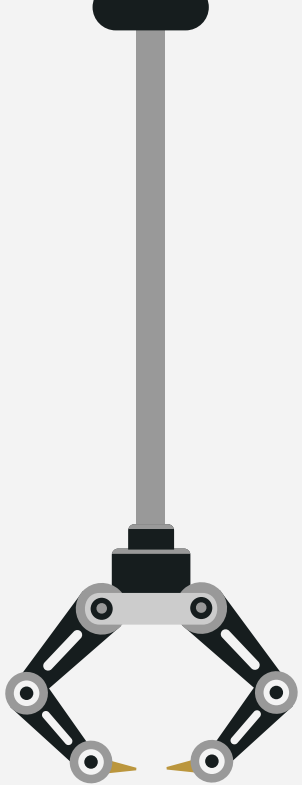
Random failures and scrap

Lever 1 size 10 to 50

Random failures and scrap

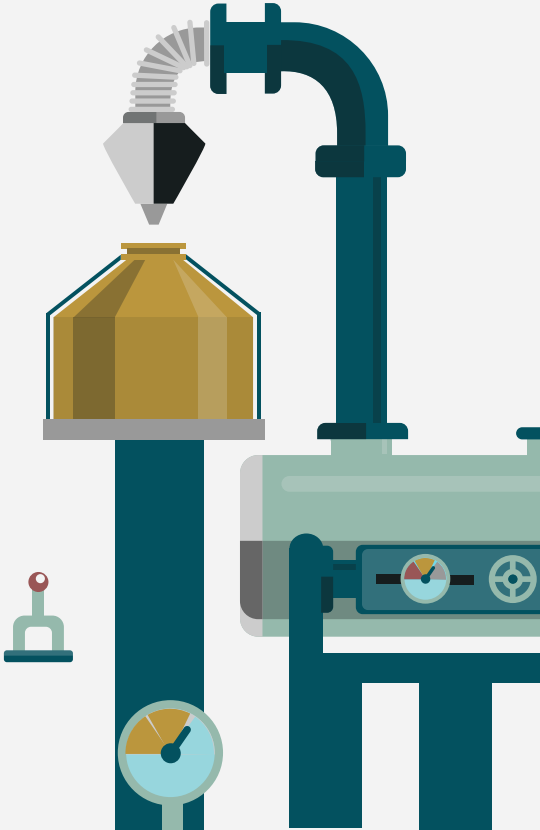
Lever 2: preventive maintenance on every machine
Reactive (no PM), or stop after 100, 150, 200 or 250 operating hours

5 sizes of B 1 × 5 sizes of B 2 × 5 policies = 125 scenarios



02

Data & Inputs





From raw data to simulation-ready inputs

Method overview — every simulated random input is traceable to workbook evidence

1) Extract
Read the source columns used for each random input.

2) Clean
Remove missing, invalid or non-modelable observations.

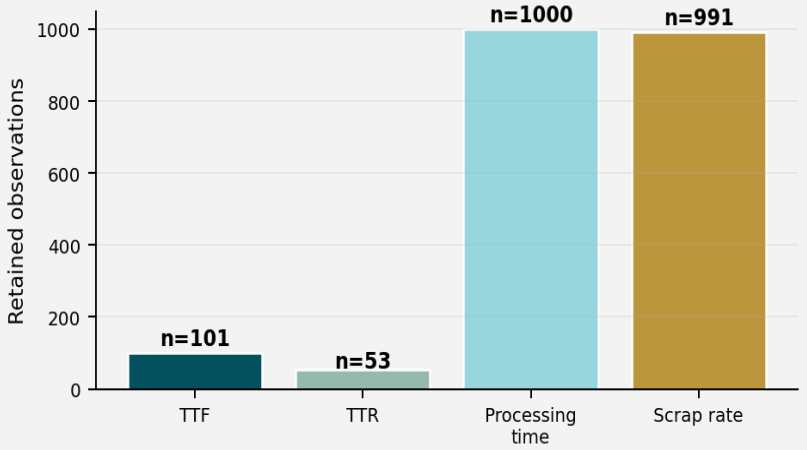
3) Fit
Estimate candidate distributions and compare fit evidence.

4) Parameterize
Write accepted parameters into Model Inputs.

5) Simulate
Sample new values during each Monte Carlo run.

Why the cleaning stage matters

- Distribution fitting is sensitive to impossible values and extreme data errors.
- The cleaning rule must be reproducible — not adjusted until a preferred distribution appears.



Data cleaning logic before distribution fitting

What was done in this cleaning step

- **TTF:** collected cycle counts for 100 engines from the mechanical / electrical history.

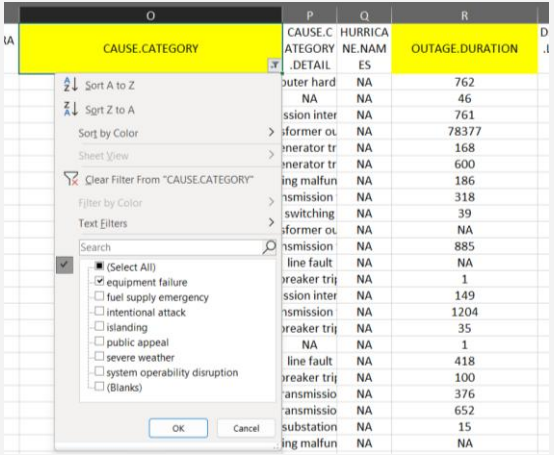
Kept one cycle value per engine and removed 0-cycle records used only for testing.

- **TTR:** started from real repair / outage data, filtered to equipment failure, removed 0, NA and non-numeric values, then converted duration from minutes to hours.
- **Processing time and scrap rate were cleaned with the same logic:** keep valid steady-state observations and remove impossible values.

Very radical values were removed only when they reflected installation, start-up or another non-steady-state period.

General rule

Keep steady-state data; remove impossible / non-representative extremes.



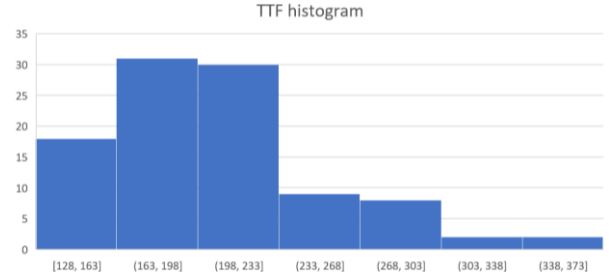
Filter cause = equipment failure
Remove 0 / NA values
Convert duration from min to h

	A	B
3	Engine_ID	TTF (Cycles)
4	1	192
5	2	287
6	3	179
7	4	189
8	5	269
9	6	188
10	7	259

100 engines, cycle values per engine

TTF distribution fitting: from histogram to candidate models

1) Explore the empirical shape



2) Fit Weibull using the cleaned TTF statistics

Statistic	Value
Number of observation	100
Mean TTF	206.31
Standard Deviation	46.34274921
Min TTF	128
Max TTF	362

Linearized Weibull equation

$$F(t) = 1 - e^{-(t/\beta)^{\alpha}}$$

$$\ln\left[-\ln\left(1 - F(t)\right)\right] = \alpha \ln(t) - \alpha \ln(\beta)$$

$$Empirical\ F_i = (i - 0.5)/n$$

Paramater	Value
shape	5.6173
scale	223.2897514
R^2	0.8877
Avr Ln	5.306244852
Std dev Ln	0.212115711

3) Build intervals and prepare the χ^2 test

Number of intervals was selected by Sturges' rule:

$$k = 1 + 3.322 \log_{10}(n).$$

For n = 100: $k = 1 + 3.322 \log_{10}(100) \approx 7\ bins.$

Expected frequency per interval:

$$E_i = n \times [F(upper_i) - F(lower_i)].$$

Goodness-of-fit statistic:

$$\chi^2 = \Sigma\,(O_i - E_i)^2 / E_i.$$

Sorted TTF I (t i)	Rank i	Emprical F(t) i	Ln(ti)	ln(ln(1/(1-F))	Weibull fitted F(x)	Exponential F(t)
128	1	0.005	4.85203	-5.29581214	0.042956617	0.462284448
135	2	0.015	4.90527	-4.19215778	0.057494729	0.480222839
137	3	0.025	4.91998	-3.67624726	0.062288618	0.485237292
137	4	0.035	4.91998	-3.33464652	0.062288618	0.485237292
147	5	0.045	4.99043	-3.07815915	0.091115044	0.509593183
147	6	0.055	4.99043	-2.87227026	0.091115044	0.509593183
150	7	0.065	5.01064	-2.69995184	0.10149037	0.516672702
153	8	0.075	5.03044	-2.55153963	0.11273323	0.52365002
154	9	0.085	5.03695	-2.42101719	0.116680019	0.525953337
154	10	0.095	5.03695	-2.30438336	0.116680019	0.525953337

TTF candidate testing and final distribution selection

Candidate A — Weibull distribution

E i	New Ei i	New O i	Chi- sqaure
10.1490488	10.1490488	6	1.69617925
15.62210712	15.62210712	21	1.8513336
24.98756478	24.98756478	35	4.01194994
26.92585062	26.92585062	19	2.3330408
16.84980537	16.84980537	8	4.64806882
4.942602617	5.46562331	11	5.60399493
0.510344595		SUM	20.1445673
0.012631771		Degrees of Freedom	3
4.43268E-05		Chi-Square critical	7.8147279

$$\chi^2\ calculated\ =\ 20.1446$$
$$\chi^2\ critical\ =\ 7.8147\ \rightarrow\ 20.1446\ >\ 7.8147$$

Conclusion: Weibull does not adequately describe the observed TTF data.

REJECTED

Candidate B — Lognormal distribution

E i	New Ei i	New O i	Chi- sqaure
8.171557937	8.171557937	6	0.577082597
21.49254872	21.49254872	21	0.01128783
27.9739782	27.9739782	35	1.764675083
21.81275967	21.81275967	19	0.362705914
12.130375	12.130375	8	1.406386666
5.373324074	8.418780472	11	0.791408479
2.037802411			
0.694511703			
0.313142283		SUM	4.913546569
		Chi-Square Critical	7.814727903

$$\chi^2\ calculated\ =\ 4.9135$$
$$\chi^2\ critical\ =\ 7.8147\ \rightarrow\ 4.9135\ <\ 7.8147$$

Conclusion: lognormal fit was retained for simulation.

NOT REJECTED

Final TTF input

Distribution: Lognormal
ln-mean μ = 5.306245
ln-standard deviation σ = 0.212116

TTR distribution fitting: repair records → Weibull candidate

1) Source data and cleaning

CAUSE.CATEGORY	OUTAGE.DURATION
equipment failure	762
equipment failure	46
equipment failure	761
equipment failure	78377
equipment failure	168
equipment failure	600
equipment failure	186
equipment failure	318
equipment failure	39
equipment failure	NA

- Filter CAUSE.CATEGORY to equipment failure
- Remove 0, NA and non-numeric outage durations
- Convert outage duration from mins to hrs
- Exclude abnormal extreme records not representing steady-state repair time

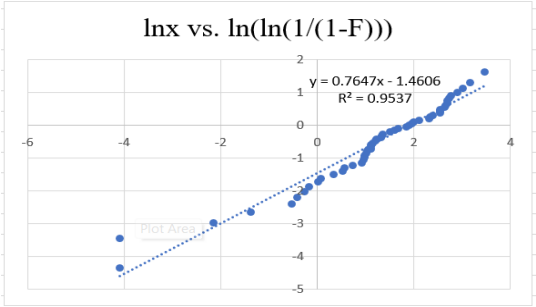
Example extreme: 1306.28 h outage was not used for the fitted steady-state sample.

2) Statistics and interval rule

Count	53
Mean	6.777673
Median	3.683333
Max	31.9
Min	0.016667
Std dev	6.872884
Mean ln	1.199049
Std dev ln	1.563606
Shape	0.764693
Intercept	-1.46056
Scale	6.753039
N of Bins	7

The same seven intervals were then used for the χ^2 goodness-of-fit check.

3) Weibull fitting evidence



Ln(TTR)	Rank i	F(ti)	Sorted TTR	ln(Sorted TTR)	Weibull Y
2.541601993	1	0.013207547	0.016666667	-4.094344562	-4.320326452
-0.265703166	2	0.032075472	0.016666667	-4.094344562	-3.423407366
2.540288796	3	0.050943396	0.116666667	-2.148434413	-2.951010633
1.029619417	4	0.069811321	0.25	-1.386294361	-2.625993381
2.302585093	5	0.088679245	0.583333333	-0.538996501	-2.376658496
1.131402111	6	0.10754717	0.65	-0.430782916	-2.173474299
1.667706821	7	0.126415094	0.766666667	-0.265703166	-2.001370358
-0.430782916	8	0.145283019	0.833333333	-0.182321557	-1.851605777
2.691243083	9	0.164150943	1.016666667	0.016529302	-1.718654539
-4.094344562	10	0.183018868	1.066666667	0.064538521	-1.598798321

TTR candidate testing and final repair-time input

Candidate distributions were checked with interval probabilities and χ^2 statistics from the workbook before selecting the simulation input.

Candidate A — Weibull distribution

BORDERLINE / RETAINED

Interval	CDF	Lower Bound	Upper bound	O i	F(Lower)	F(Upper)	Pi	E i	Chi- sqaure
1	0.14	0	0.585549848	5	0	0.142857073	0.14286	7.571424849	0.873313265
2	0.29	0.585549848	1.625114593	6	0.142857073	0.285714217	0.14286	7.571428638	0.326145577
3	0.43	1.625114593	3.16090426	13	0.285714217	0.428571388	0.14286	7.571430064	3.892180381
4	0.57	3.16090426	5.437397044	8	0.428571388	0.571428572	0.14286	7.571430779	0.024258503
5	0.71	5.437397044	9.067422056	5	0.571428572	0.714285759	0.14286	7.571430895	0.873316674
6	0.86	9.067422056	16.12832502	12	0.714285759	0.85714293	0.14286	7.571430075	2.590294222
7	1.00	16.12832502	infinity	4	0.85714293		0.14286	7.571424701	1.684633328
								SUM	10.26414195
								Chi-Square Critical	9.487729037

Candidate B — Lognormal distribution

REJECTED

Interval	CDF	Lower Bound	Upper bound	O i	F(Lower)	F(Upper)	Pi	E i	Chi- sqaure
1	0.14	0	0.585549848	5	0	0.133685534	0.13369	7.085333289	0.613748817
2	0.29	0.585549848	1.625114593	6	0.133685534	0.324087675	0.1904	10.09131348	1.658738083
3	0.43	1.625114593	3.16090426	13	0.324087675	0.487706311	0.16362	8.671787711	2.160272166
4	0.57	3.16090426	5.437397044	8	0.487706311	0.62403553	0.13633	7.225448639	0.083030112
5	0.71	5.437397044	9.067422056	5	0.62403553	0.739937729	0.1159	6.14281653	0.212610879
6	0.86	9.067422056	16.12832502	12	0.739937729	0.844102253	0.10416	5.520719778	7.604275146
7	1.00	16.12832502	infinity	4	0.844102253		0.1559	8.262580575	2.199021601
								SUM	14.5316968
								Chi-Square Critical	9.487729037

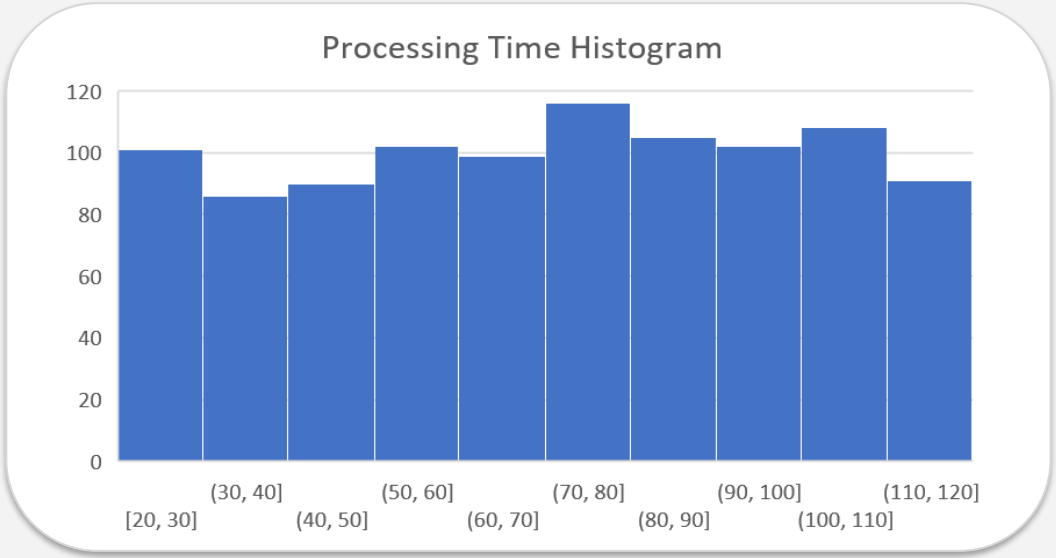
Final TTR simulation input

TTR ~ Weibull(α, β)

- shape α = 0.764693
- scale β = 6.753039 h
- unit in simulation: hours
- cleaned sample used for fitting: n = 53

Processing time distribution fitting: uniform model

The cycle-time input was modeled with a bounded uniform distribution, not with Sturges-based bins.



Data used

- Processing time per unit.
- Valid numeric cycle times only.
- n = 1,000 observations.
- Observed range: 20–120 s/unit.

Count	1000
Mean	71.384
Median	72
Max	120
Min	20
Std.dev	28.4844

N of intervals	11
Bin Width	9.09091

$$\textit{Uniform candidate: } X \sim U(a,b), \quad a = 20, b = 120 \rightarrow f(x) = 1/(b-a), 20 \leq x \leq 120$$

Processing time goodness-of-fit and final simulation input

Equal expected frequencies were used across the uniform intervals before retaining the final model.

Lower Bound	Upper Bound	O i	F(Lower)	F(Upper)	P i	E i	Chi-square
20	29.09090909	92	0	0.090909	0.090909	90.90909091	0.01309091
29.09090909	38.18181818	75	0.090909	0.181818	0.090909	90.90909091	2.78409091
38.18181818	47.27272727	82	0.181818	0.272727	0.090909	90.90909091	0.87309091
47.27272727	56.36363636	84	0.272727	0.363636	0.090909	90.90909091	0.52509091
56.36363636	65.45454545	94	0.363636	0.454545	0.090909	90.90909091	0.10509091
65.45454545	74.54545455	98	0.454545	0.545455	0.090909	90.90909091	0.55309091
74.54545455	83.63636364	92	0.545455	0.636364	0.090909	90.90909091	0.01309091
83.63636364	92.72727273	96	0.636364	0.727273	0.090909	90.90909091	0.28509091
92.72727273	101.8181818	96	0.727273	0.818182	0.090909	90.90909091	0.28509091
101.8181818	110.9090909	100	0.818182	0.909091	0.090909	90.90909091	0.90909091
110.9090909	120	91	0.909091	1	0.090909	90.90909091	9.0909E-05
	SUM	1000				SUM	6.346
						Degress of Freedom	8
						Chi-sqaure critical	15.5073131

The null hypothesis is not rejected at α = 0.05; therefore, the Uniform distribution provides an acceptable fit to the processing-time data.

Interval setup

- Range: 20–120 s/unit.
- Same expected frequency in each interval.
- $E_i = n/k = 1000/11 = 90.909$.

Goodness of fit

- χ^2 calculated = 6.346
- df = 11 − 2 − 1 = 8
- χ^2 critical = 15.507 at α = 0.05

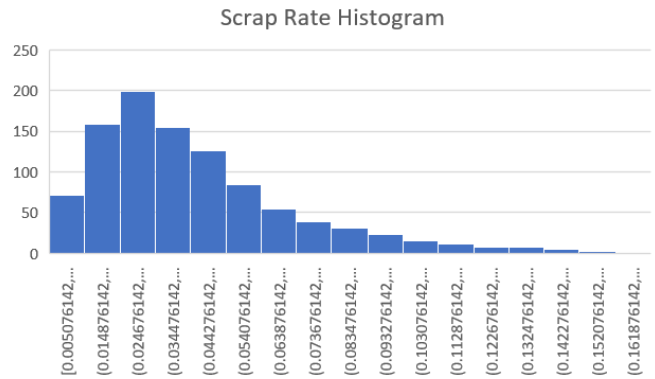
Decision Since 6.346 < 15.507, the null hypothesis is not rejected at α = 0.05. The Uniform(20,120) model is retained.

Final simulation parameterization

- Processing time ~ Uniform(20,120); unit used in simulation: seconds per unit.

Scrap rate distribution fitting: right-skewed quality-loss input

The histogram showed a positive, right-skewed pattern, so a lognormal candidate was used for the scrap-rate model.



Count	991
Mean	0.04529
Median	0.03774
Max	0.16216
Min	0.00508
Std.dev	0.02802
Avr Ln	-3.277
Std.dev Ln	0.61881

Candidate model

- Lognormal
- Input remains a fraction

Interval setup

- Equal-probability intervals used.
- Each class has Pi = 0.10.
- Ei = 991×0.10 = 99.1.

Scrap rate goodness-of-fit and final simulation input

Decile intervals were used for the χ^2 test; the fitted lognormal distribution was retained at the 5% significance level.

Prob	Lower	Upper	O i	Pi	E i	Chi-square
0.1	0	0.01708	104	0.1	99.1	0.2422805
0.2	0.01708	0.02242	86	0.1	99.1	1.7316852
0.3	0.02242	0.02728	94	0.1	99.1	0.2624622
0.4	0.02728	0.03227	109	0.1	99.1	0.989001
0.5	0.03227	0.03774	103	0.1	99.1	0.1534813
0.6	0.03774	0.04415	84	0.1	99.1	2.3008073
0.7	0.04415	0.05221	118	0.1	99.1	3.6045409
0.8	0.05221	0.06353	94	0.1	99.1	0.2624622
0.9	0.06353	0.08341	95	0.1	99.1	0.1696266
1	0.08341		104	0.1	99.1	0.2422805
		SUM	991		SUM	9.9586276
					Degrees of freedom	7
					Critical Chi-sqaure	14.06714

Workbook χ^2 goodness-of-fit table using equal expected frequencies.

$$\chi^2 = \Sigma\,(Oi - Ei)^2 / Ei$$

df = k − p − 1 = 10 − 2 − 1 = 7

Goodness of fit

- χ^2 calculated = 9.9586
- χ^2 critical = 14.067
- α = 0.05, df = 7

Decision

- 9.9586 < 14.067.
- Null hypothesis is not rejected.
- Lognormal fit is acceptable.

Final simulation parameterization

- $Scrap\ rate \sim Lognormal(\mu = -3.277, \sigma = 0.61881)$.

Modeling note

- B ased on cleaned steady-state data.
.

Model Inputs — machine-specific parameterization

Adjustment logic: machine input = base fitted distribution × multiplier; only machine-specific parameters are changed, not the distribution family.

Time to Failure (TTF)

Machine	Dist.	Multiplier	Adjusted μ	Mean TTF (h)
M1	Lognormal	1.0	5.306245	206.18
M2	Lognormal	0.8	5.083101	164.94
M3	Lognormal	1.2	5.488567	247.41

Time to Repair (TTR)

Machine	Dist.	Multiplier	Adjusted η	Mean TTR (h)
M1	Weibull	1.0	6.7530	7.92
M2	Weibull	0.8	5.4024	6.33
M3	Weibull	1.2	8.1036	9.50

Processing Time (PT)

Machine	Dist.	Multiplier	Adjusted min	Adjusted max	Mean PT (s)
M1	Uniform	1.0	20	120	70
M2	Uniform	0.8	16	96	56
M3	Uniform	1.2	24	144	84

Scrap Rate

Machine	Dist.	Multiplier	Adjusted μ	Mean scrap rate
M1	Lognormal	1.0	−3.27697	0.04571
M2	Lognormal	1.0	−3.27697	0.04571
M3	Lognormal	1.0	−3.27697	0.04571

These parameter tables are the final bridge between distribution fitting and the discrete-event Monte Carlo simulation logic.

Simulation settings — decision variables and baseline scenario

The fitted random inputs are combined with controllable PM and buffer decisions before running the Monte Carlo model.

Parameter	Type	B aseline	Alternative scenarios	Unit / note
PM interval	Decision variable	200	Reactive, 100, 150, 200, 250	hours
PM duration	Assumption	1	Fixed	hour/event
B uffer 1 capacity	Decision variable	30	10, 20, 30, 40, 50	units
B uffer 2 capacity	Decision variable	30	10, 20, 30, 40, 50	units
Demand target	Scenario input	300	Fixed initially	good units/shift
Shift duration	Model setting	8	Fixed	hours

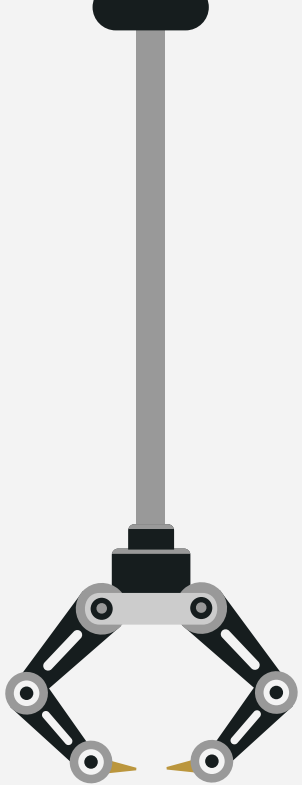
Baseline experiment

PM interval = 200 h; PM duration = 1 h/event.
B 1 = B 2 = 30 units; demand = 275.

Scenario sweep

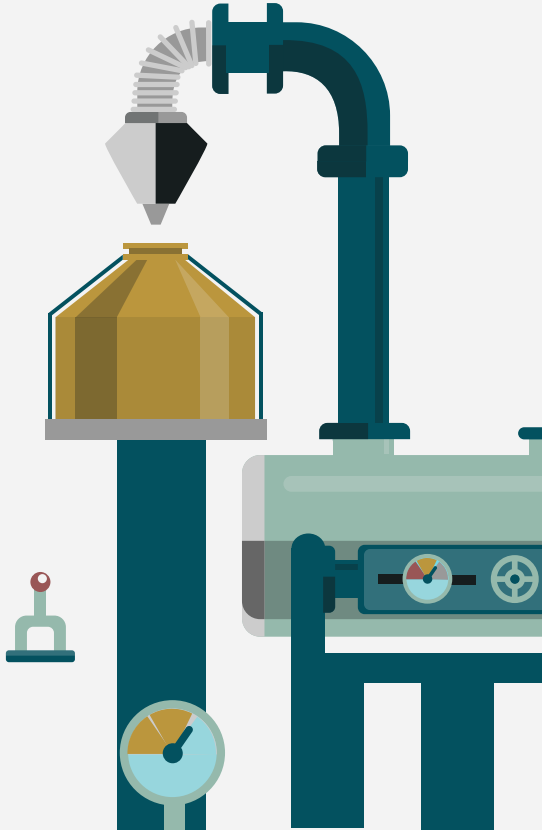
Compare PM policy and buffer capacity combinations.
Reactive is the no-PM reference.

Experimental design: PM policy × buffer capacity × machine behavior → throughput, downtime, scrap and demand achievement.



03

Monte Carlo in Python





Actual machine-start logic used in Python

Executed downstream-first at every event cycle to release buffer capacity before upstream starts.

M3 — starts first

```
if m3.state == "IDLE" and b2 >
0:
    b2 -= 1
    start_processing(m3, rng)
```

M2 — starts second

```
if m2.state == "IDLE" and b1 >
0
and b2 < buffer2_capacity:
    b1 -= 1
    start_processing(m2, rng)
```

M1 — starts last

```
if m1.state == "IDLE"
and b1 < buffer1_capacity:
    start_processing(m1, rng)
```

M1 → B1 → M2 → B2 → M3 → Finished Good

Blocking condition

M1: B 1 full
M2: B 2 full

Starvation condition

M2: B 1 empty
M3: B 2 empty

Material movement

M2 start: B 1 − 1
M3 start: B 2 − 1



Event-driven simulation clock

The model does not advance in fixed time steps; it jumps to the next event.

Candidate event times

- End of 720 h horizon
- remaining processing/repair/PM
- time until next failure
- time until next PM

Advance time

```
dt = min(candidates)

t += dt
```

Update each machine

```
PROCESSING → age + dt
REPAIR → repair time + dt
PM → PM time + dt
IDLE → blocked/starved/idle
```

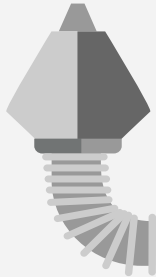
Only PROCESSING increases operating_age_h.

Failure candidate

```
next_failure_age_h
– operating_age_h
```

PM candidate

```
pm_interval_h
– operating_age_h
```



Failure, PM and reset logic

1 — Failure has priority

```
if PROCESSING and age >= next_failure_age:
    state = "REPAIR"
    remaining_h = sample_ttr(...)
    failures += 1
    continue
```

2 — Otherwise check PM

```
if PROCESSING and PM is active
and age >= pm_interval:
    state = "PM"
    remaining_h = pm_duration
    pm_events += 1
```

Repair / PM completion

```
if remaining_h <= eps:
    reset_after_maintenance(m, rng)
```

Reset used in the model

```
operating_age_h = 0
next_failure_age_h = sample_ttf(...)
state = "IDLE"
remaining_h = 0
```

Model assumption: an interrupted unit is lost when failure/PM occurs; it is not resumed.

From one run to the Monte Carlo experiment

1 Configure

Choose B 1, B 2 and PM policy

2 Simulate

Run the line for 720 h (1 month)

3 Record

Finished goods, downtime, failures…

4 Repeat

1,000 runs for the same scenario

Design space

5 B 1 values × 5 B 2 values × 5 PM strategies

125

configurations

125,000

simulated months



What happens inside one Monte Carlo run

Initialize

Seed RNG
Sample initial TTF
B 1 =B 2 =FG=0

Start eligible machines

M3 → M2 → M1
using IF conditions

Find next event

dt = minimum
candidate time

Update states

Processing / repair
/
PM / blocked /
starved

Resolve event

Failure → PM →
completion/scrap

Repeat until t = 720 h

Outputs saved after each run

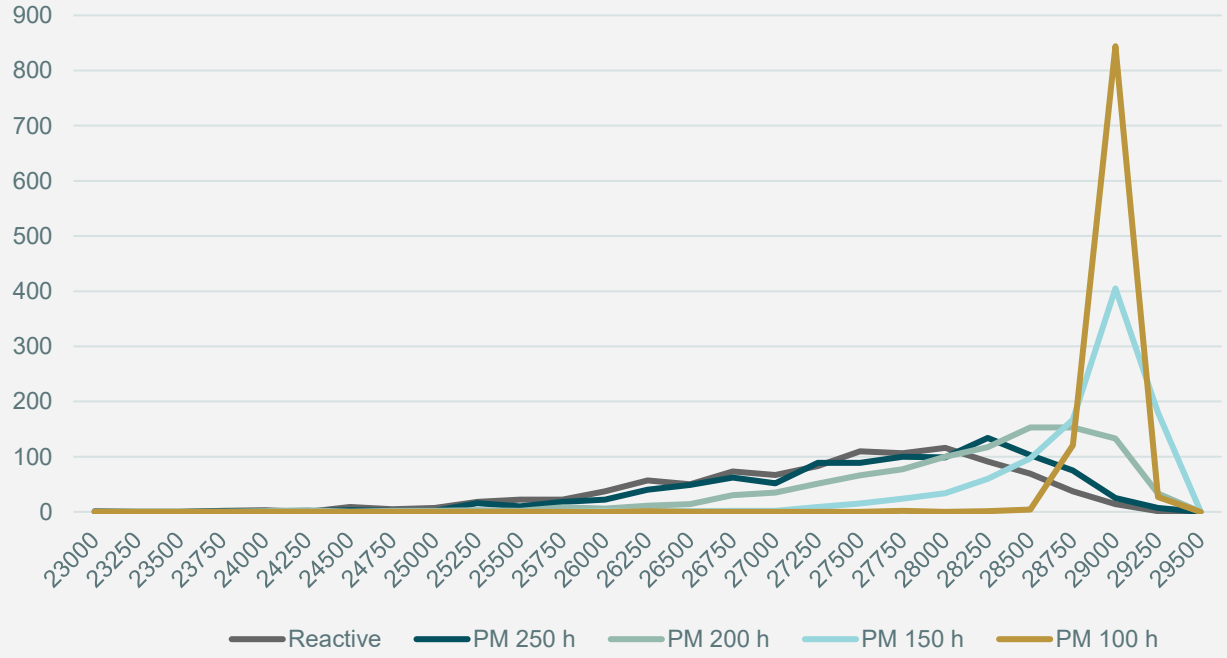
Finished Good • Demand Met • Throughput • Failures • PM Events • Repair/PM Time • Blocking • Starvation • Availability

From code execution to output distributions

Scenario grid
5 B 1 × 5 B 2 × 5 PM
= 125 configurations

Replication
1,000 runs
per configuration

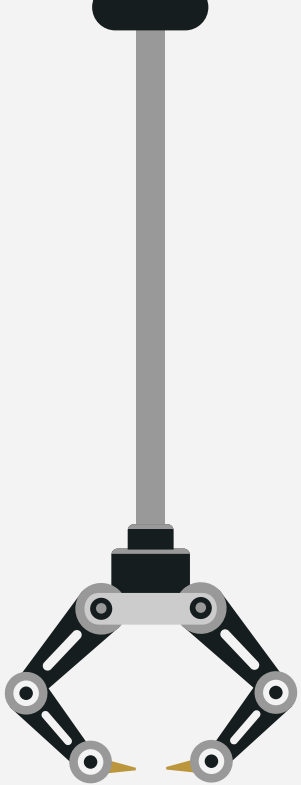
Main experiment
125,000 simulated
production months



Focused distribution
B 1 = B 2 = 30

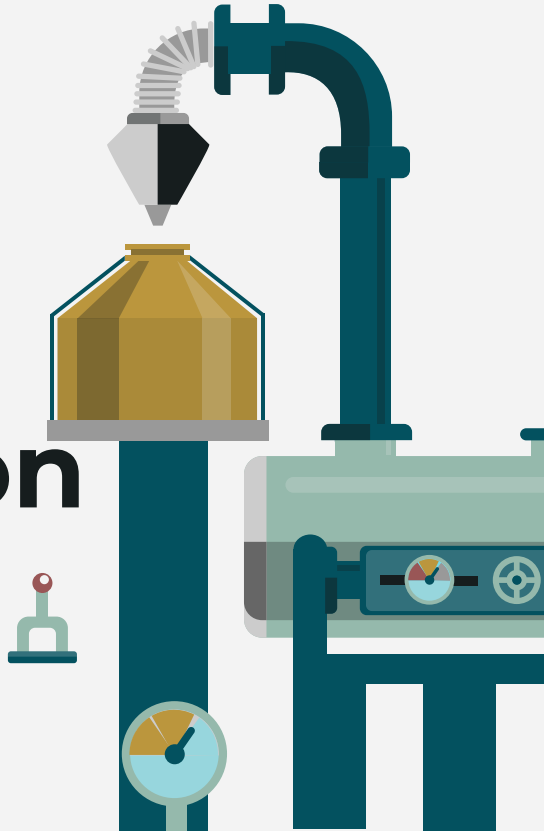
Raw observations
5 PM × 1,000 = 5,000 runs

Purpose
Variability + demand risk



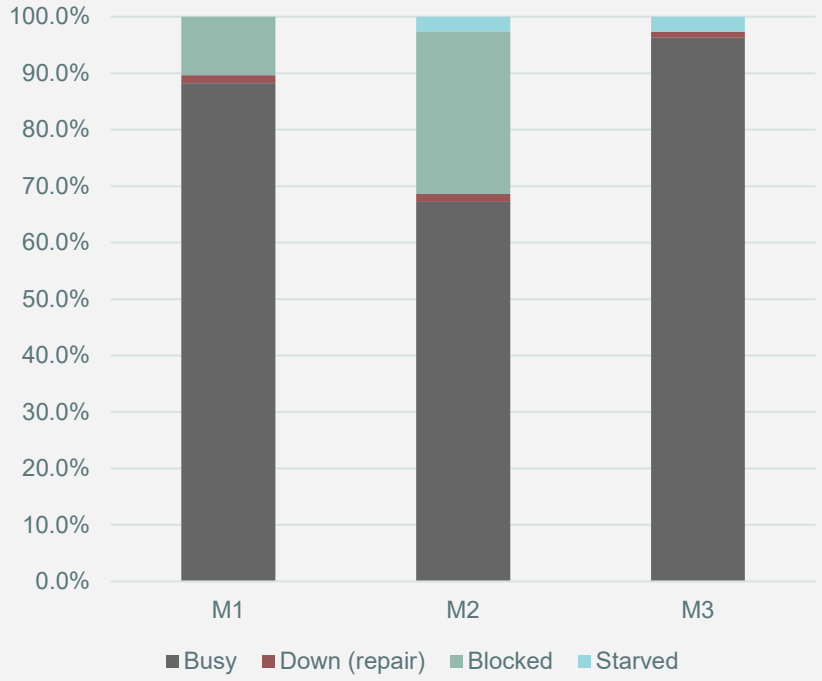
04

Analysis & Recommendation





M3 is the bottleneck: busy 96% of the time



Utilization = (total time − downtime − blocked − starved) / total time; downtime = repair only.

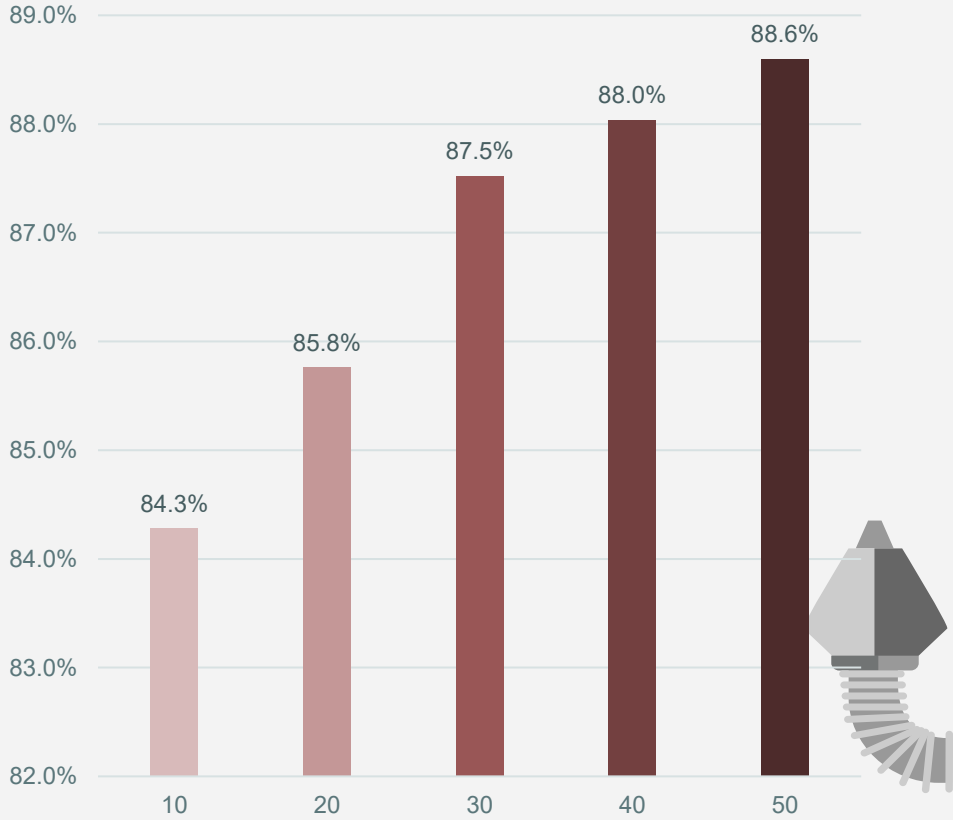
Key takeaways:

- M3 is busy 96% of the time. It almost never waits, and it has no blocked time because nothing is downstream of it.
- M2 is busy only 67%. It loses 29% of its time blocked, because B2 fills up when M3 can't keep pace.
- Breakdowns are small on every machine (about 1% each). The lost time is waiting, not repairs.
- M3 has the highest utilization in all scenarios.
- Improving or protecting M3 raises line output. Speeding up M1 or M2 won't.

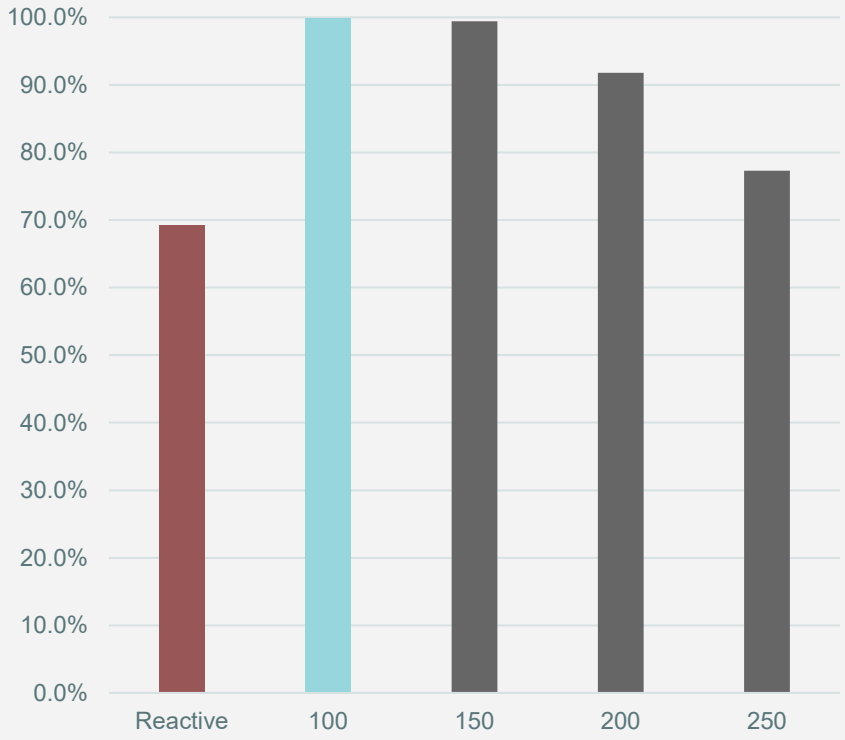
Buffer gains flatten after 30

Key takeaways:

- Going from 10 to 30 adds 3.2 points of demand-meeting probability.
- Going from 30 to 50 adds only 1.1 more.
- Beyond 30, extra buffer space is mostly wasted cost.
- Takeaway: A buffer of about 30 is the practical sweet spot.



PM every 100h: better and more consistent



Key takeaways:

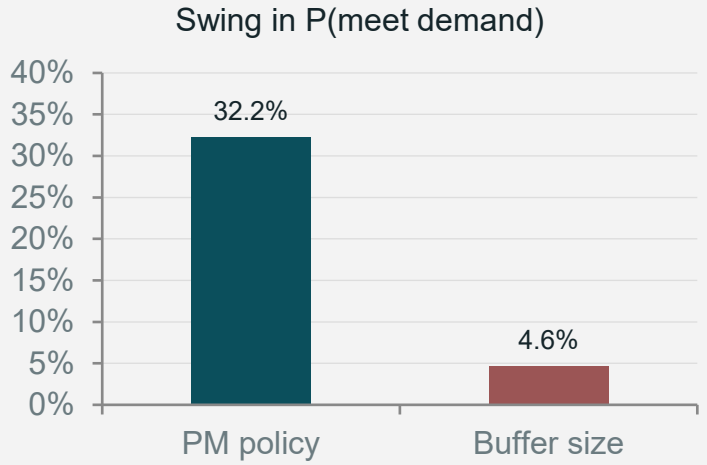
- PM 100h meets demand in 99.9% of runs vs 69.2% for Reactive.
- Mean output rises from 27,422 to 29,088 units (+6%).
- Variability falls 86% (SD 991 to 141), so the line is predictable.
- Benefits fade as PM gets rarer: 250h is only 77%.
- Takeaway: frequent PM raises output and makes it reliable.

Maintenance dominates; buffer barely matters

Key takeaways:

- Sensitivity: PM swings P(meet demand) by 32.2 pp, buffer by only 4.6 pp.
- Regression: Reactive to PM 100h is worth +31.8 pp; +10 buffer slots is worth about +0.5 pp.
- B 2 helps slightly more than B 1.
- With PM 100h, buffer changes the result by only 0.1 percentage point (pp).

Factor (vs Reactive)	Effect	p-value
PM 100h	+31.8 pp	<0.001
PM 150h	+30.7 pp	<0.001
PM 200h	+22.2 pp	<0.001
PM 250h	+9.9 pp	<0.001
B1, per +10 slots	+0.4 pp	<0.001
B2, per +10 slots	+0.5 pp	<0.001



The top scenarios are statistically tied

Scenario (B1, B2, PM)	Mean output	95% CI	P(meet)
50, 50, 100h	29,146.2	+/- 6.9	100%
20, 50, 100h	29,141.6	+/- 7.1	100%
30, 50, 100h	29,140.8	+/- 8.3	99.9%
10, 50, 100h	29,139.4	+/- 6.6	100%
40, 50, 100h	29,137.6	+/- 8.3	100%
30, 30, 100h (our pick)	29,088.0	+/- 8.7	99.9%

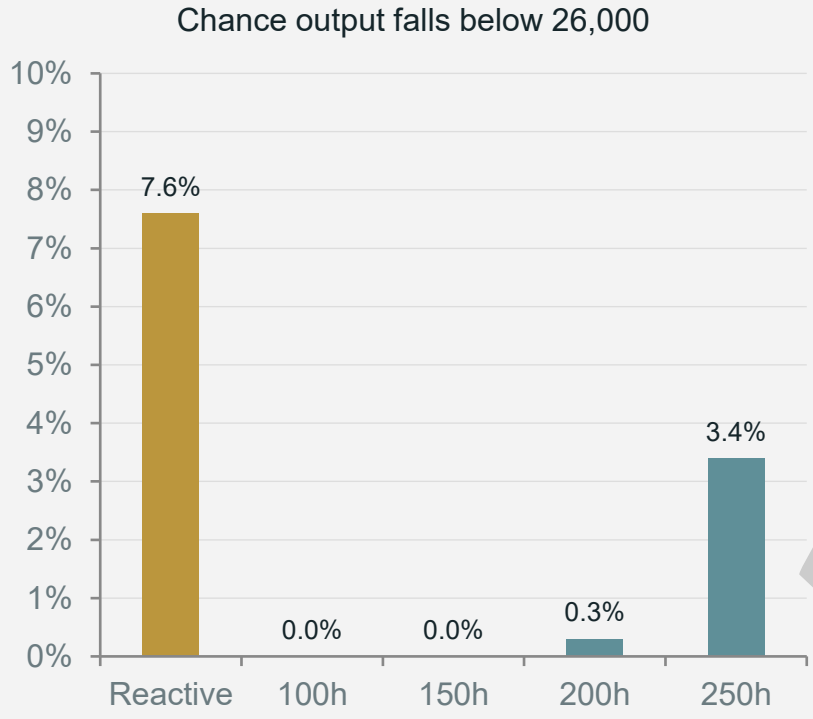
Key takeaways:

- The top five scenarios all use PM 100h with B 2 = 50, and their intervals overlap.
- They span under 10 units (0.03%), so we cannot rank them.
- O ur pick (30, 30, 100h) is 58 units (0.2%) lower, which is negligible: it still meets demand 99.9% of the time.
- Do not claim one exact buffer is best. The PM finding is what is solid.

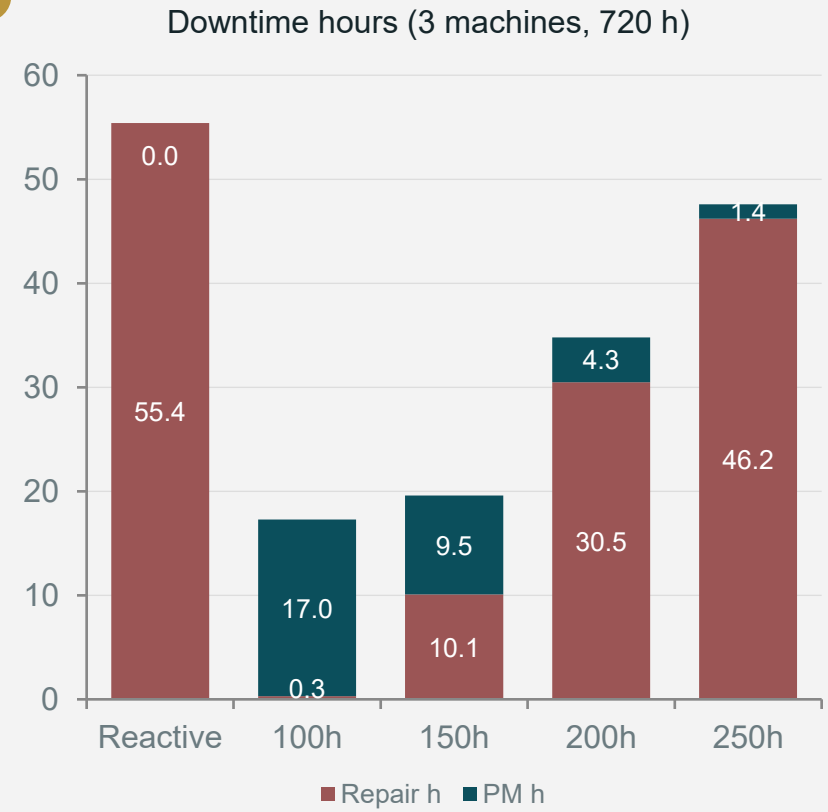
Reactive maintenance risks serious shortfalls

Key takeaways:

- Bad month = output below 26,000 (about 4% under demand).
- Reactive: 7.6% chance. PM 250h: 3.4% . PM 200h: 0.3% . PM 100h and 150h: about 0% .
- Example: z = (26,000 - 27,422) / 991 = -1.43, so P = 7.6% .
- Reactive output is skewed toward bad months, so the true risk is higher: 8.9% in the actual runs.



Scrap is flat; PM cuts downtime by 69%



Key takeaways:

- Scrap rate is 13.10% to 13.12% in all 125 scenarios, so quality does not confound the results (scrap is constant by model assumption).
- Total downtime falls from 55.4 h (Reactive) to 17.4 h (PM 100h).
- PM does not just swap unplanned for planned time: the total really drops, and what remains is schedulable.

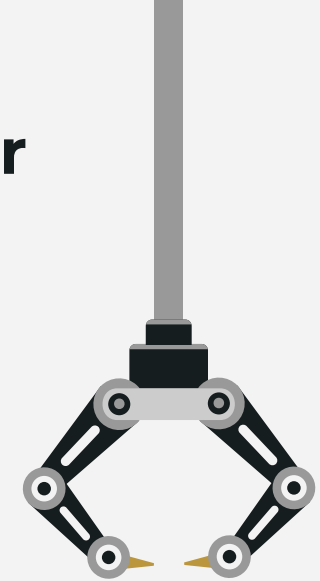
Recommendation: PM every 100h, buffer about 30

Bottleneck

M3, busy 96% of the time. M2 loses 29% of its time blocked, waiting on M3.

What matters most

Maintenance (32 pp swing) far more than buffer size (5 pp).



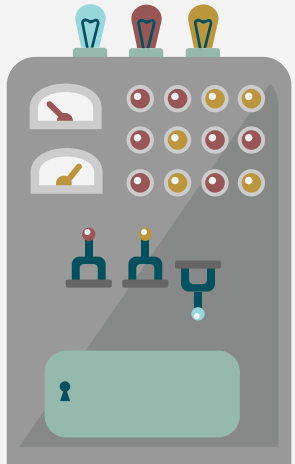


Setting

PM every 100h: 99.9% chance of meeting demand, 86% lower variability, 69% less downtime than Reactive.

Buffer

About 30, where gains flatten. With PM 100h even small buffers work, so buffer is insurance.



Thank you for your attention!

Do you have any questions?

